

sTMS Interface Functional Specification

Temasek Polytechnic – Course Website

Version 1.0

Document Version

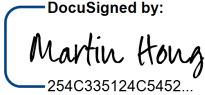
Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	22/03/2022	Initial version of TP Course Website IFS	AvePoint
0.2	05/04/2022	Revised based on feedback from TP: Update TP approver; Add Course type to 3. Interface fields and 5. Interface Schemas; Update Sample data.	AvePoint
0.3	12/04/2022	Revised based on feedback from TP: 2.2. Interface Design Summary, Add file retention policy, file transfer policy. 3. Interface Fields: Add TP website mapping fields. 3.4. Course schedule: Update fields. 4.1. Assumption: Change the description. 4.2. Constraints: Add interface trigger point. 5. Interface Schemas: Update Sample data. Add 6. Interface Fields Mapping List.	AvePoint
0.4	25/04/2022	Revised based on feedback from TP: Update 2.2. Interface Design Summary D= Deactivate. 4.2. Constraints. Update No.4; Add No.5 to clarify unmapped course data.	AvePoint
1.0	05/05/2022	Official version for acceptance and sign-off	AvePoint

Document Acceptance Form

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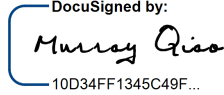
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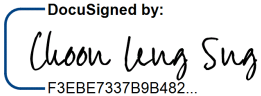
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Date:

Table of Contents

Document Version	1
Document Acceptance Form	2
1. Introduction	4
1.1 Purpose of this document	4
1.2 Intended Audience	4
2. Technical Overview	5
2.1 System Flow	5
2.2 Interface Design Summary	6
2.3 Business Flow	7
3. Interface Fields	8
3.1 Common Interface Fields	8
3.2 About Course Fields	8
3.3 Course Structure Fields	10
3.4 Course Schedule Fields	12
3.5 Entry Requirements Fields	14
3.6 Financial Matters Fields	16
3.7 Additional Information Fields	18
4. Assumption and Constraints	19
4.1 Assumption	19
4.2 Constraints	19
5. Interface Schemas	20
6. Interface Fields Mapping List	28

1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Temasek Polytechnic Website for publishing CET course and schedule information.

sTMS shall identify courses with UI changes, generate corresponding JSON files for these courses, and deposit these JSON files and course images encrypted by PGP protocol in the SFTP server accordingly. Temasek Polytechnic can update its website with the latest course information.

1.2 Intended Audience

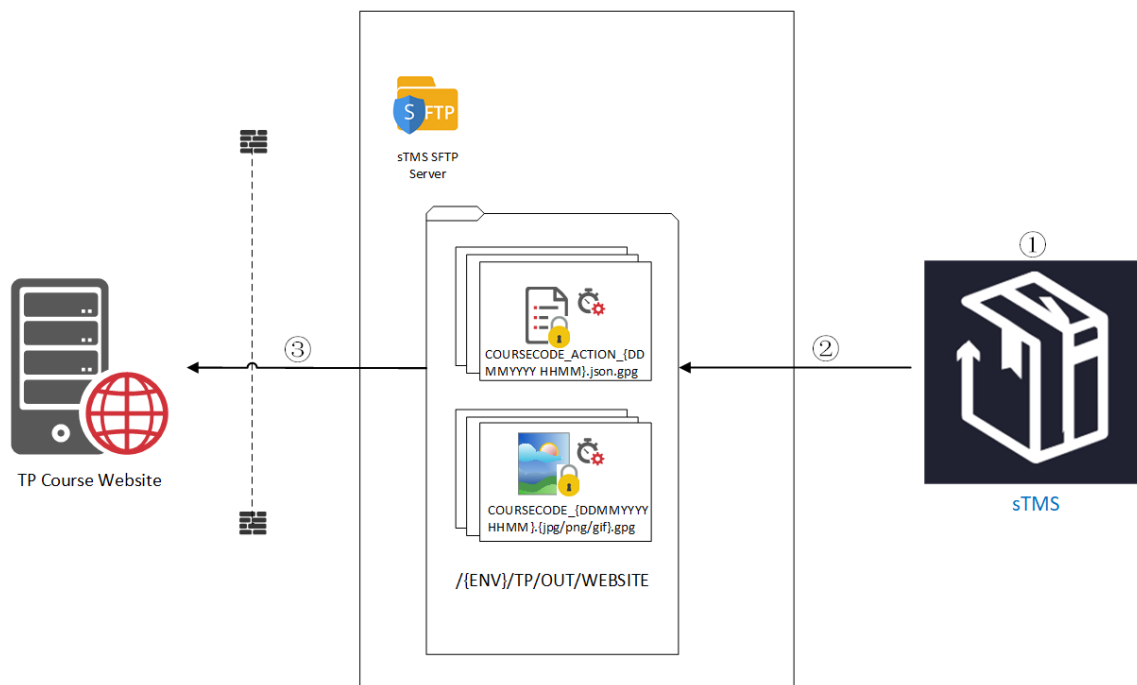
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **TP Course Website:** Temasek Polytechnic course website.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted JSON files used for data exchange.
- **sTMS:** shared Training Management System.



SN	Description
1	sTMS generates these course info files based on institution, course status, and modified time on a pre-defined schedule.
2	sTMS configures Timer job to push course info JSON files and course images (Images are uploaded by CET user from course catalogue: About course) into sTMS SFTP server after encryption with PGP protocol. For timer job trigger point and file generation rules can refer to 4.2. Constraints .
3	TP pulls these course info files and images from SFTP on demand.

2.2 Interface Design Summary

Output Interface File

Interface File Name	COURSECODE_ACTION_{DDMMYYYY HHMM}.json.gpg COURSECODE_{DDMMYYYY HHMM}.{jpg/png/gif}.gpg
Data Source System	shared Training Management System
Data Destination System	Temasek Polytechnic Course Website
Mode of Interface	JSON file, jpg/png/gif file
Method	SFTP
File Path	/ENV/TP/OUT/WEBSITE ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Mode	First-time full data sync + Subsequent data sync Filter by • institution = TP, • course status = active or inactive(action = D means upon course is deactivated in sTMS course catalogue will be sync to TP website to unpublish the course), • modified time later than last job time
Frequency	Sync every 4 hours (configurable)
Timing	00:00am; 04:00am; 08:00am; 12:00pm; 16:00pm; 20:00pm (configurable)
File Retention Policy	Files are retained for a maximum of 30 days in the SFTP server, after this period the files will be automatically deleted. TP can delete the files after pulling out
File Transfer Policy	sTMS shall send an empty file (Sample JSON file format is: EMPTY_{DDMMYYYY HHMM}.json) if all TP courses and its data fields don't have UI changes and have been published
Possible Exception and Exception Handling	1. File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer files could not being found in SFTP: ○ AP redeposits the files in SFTP. 2. Data error.

	Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs and files to troubleshoot the cause of the failure (data error types are listed below). ○ Field exceeds max length allowed ○ Field is mandatory but missing ○ Field is corrupted • AP redeposits the correct files in SFTP.
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2.3 Business Flow

Flow to generate Course Info JSON files and images

SN	Description
1	sTMS checks the course info modified time and locates the corresponding TP courses every 4 hours.
2	sTMS identifies courses with changes (delta changes) in course info based on their modified time and generates corresponding JSON files for these courses.
3	sTMS deposits these JSON files and images encrypted by PGP protocol in the SFTP server accordingly.

3. Interface Fields

3.1 Common Interface Fields

Name	Description	Data Type
Action	C – Create; U – Update; D - deactivate	String(\$nvarchar(1))
CourseCode	Course code	String(\$nvarchar(20))
CourseName	Course name	String(\$nvarchar(60))
ModifiedTime	Record modified time	String(\$DD-MM-YYYY hh:mm:ss)
CourseType	Course type	String(\$nvarchar(60))

3.2 About Course Fields

Name	Description	Data Type	TP Website Fields
Objective	Course objective	String(\$nvarchar(max))	Course Overview
Description	Course description	String(\$nvarchar(max))	
TargetAuds	Specifies the target audience of the course	String(\$nvarchar(max))	Who Should Attend
AdditionalInfo	Course additional information may contain the following information: Future Progression Career Opportunity General Intake Info (e.g. 2 intakes per year) Testimonial Trainer profile	String(\$nvarchar(max))	NA

3.3 Course Structure Fields

Name	Description	Data Type	TP Website Fields
CertificateDetail	Details of course certificate	String(\$nvarchar(max))	What You'll Learn
Certificates	Contains one or more certificates	Object	
AssessmentType	Type of assessment that students will be expected to complete for the course	String(\$nvarchar(max))	Modes of Assessment
CourseStructure.Certificates			What You'll Learn
CertificateName	Certificate name	String(\$nvarchar(60))	
Description	Certificate description	String(\$nvarchar(max))	
Modules	Contains one or more modules	Object	
CourseStructure.Certificate.Modules			
ModuleCode	Module Code	String(\$nvarchar(20))	
ModuleName	Module name	String(\$nvarchar(60))	
Description	Module description	String(\$nvarchar(max))	

Course structure

Course certificate

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Post Diploma Certificate 1

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Post Diploma Certificate 1



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[BM0001]e-Business Management	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...
Digital Development with Python	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...
Big data analytics and Optimisation	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...

Post Diploma Certificate 2

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Post Diploma Certificate 2



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Assessment type

Project

3.4 Course Schedule Fields

Name	Description	Data Type	TP Website Fields
Description	Course schedule description	String(\$nvarchar(max))	Course Schedule
Intakes	Contains one or more course intake	Object	
CourseSchedule.Intakes			
DurationStartDate	Course registration duration start date	String(\$DD-MM-YYYY)	
DurationEndDate	Course registration duration end date	String(\$DD-MM-YYYY)	
CourseStartDate	Course start date	String(\$DD-MM-YYYY)	
CourseEndDate	Course end date	String(\$DD-MM-YYYY)	
ScheduleInfo	Course schedule info description	String(\$nvarchar(max))	
RegistrationPriority	Applicants fall under below priority group can enjoy an earlier registration start date.	Object	
ModeOfTraining	Mode of training info	String(\$nvarchar(60))	
Venue	Venue info	String(\$nvarchar(max))	
Apply	Course application URL	String(\$nvarchar(255))	Apply
CourseSchedule.Intakes.RegistrationPriority			Course Schedule
GroupType	Priority group type. E.g: On-going students; SG citizen; SG PR; etc.	String(\$nvarchar(60))	
RegistrationStartDate	Earlier registration start date	String(\$DD-MM-YYYY)	

Course schedule

Course Schedule

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Course intake 1

[View class period](#)

Course intake No.	202101001	Registration duration	11/06/2021 - 01/07/2021
Course duration	24/10/2021 - 31/10/2022	Contact person	Ashley Lim
Email	Ashley@email.com	Phone	+6512465678

Schedule info

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Group type	Registration start date
On-going students	12/01/2021
SG citizen	23/02/2021
SG PR	25/03/2021
SG citizen, on-going student	27/03/2021
SG PR, on-going student	12/04/2021
Alumni	21/05/2021

 The intake registration is currently full, new application will be put under waitlist.

[Apply](#)

Course intake 2

[View class period](#)

Course intake No.	202101001	Registration duration	11/06/2021 - 01/07/2021
Course duration	24/10/2021 - 31/10/2022	Contact person	Ashley Lim
Email	Ashley@email.com	Phone	+6512465678

Schedule info

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⚠ The intake is full, no more application is accepted. Interested applicants can register interest for the course, we will notify you when there is new intake open for registration

[Apply](#)
[Interested](#)

Course intake 3

[View class period](#)

Course intake No.	202101001	Registration duration	11/06/2021 - 01/07/2021
Course duration	24/10/2021 - 31/10/2022	Contact person	Ashley Lim
Email	Ashley@email.com	Phone	+6512465678

Schedule info

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[Apply](#)

3.5 Entry Requirements Fields

Name	Description	Data Type	TP Website Fields
EntryRequirements	Entry requirement description	String(\$nvarchar(max))	Entry Requirements

Entry requirements

Condition 1

Highest qualification

Post-Secondary; GCE A Level; H1 Gujarati; Minimum grade: A; Graduate within 5 years.

And

Certification

Big data analysis

And

Work experience

Full time; Fabricated Metal Products, Machinery & Equipment; 3 years

Or

Condition 2

Highest qualification

Post-Secondary; GCE A Level; H1 Gujarati; Minimum grade: A; Graduate within 5 years.

And

Work experience

Part time; Electronic, Computer & Optical Products; 5 years

And

Color blindness

Pass

Remarks

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3.6 Financial Matters Fields

Name	Description	Data Type	TP Website Fields
FeePayables	Course fee payable info	Object	Course Fees
GST	Informs the applicant on whether GST is inclusive or exclusive	String(\$nvarchar(max))	NA
AllowPaymentBy	Details of eligible grants and payment methods available for the course	String(\$nvarchar(60))	NA
OtherFee	Any other supplementary fees not included in course fees	String(\$nvarchar(max))	NA
Policy	Details of refund and withdrawal policy	String(\$nvarchar(max))	NA
FundingScheme	Details of the funding schemes will be shown here, if any funding scheme is available	String(\$nvarchar(max))	NA
FinancialMatter.FeePayables			Course Fees
FeeGroupType	Fee payable group type. E.g: MES (Singapore Citizen aged 40 years and above); SC; SPR; SC SME; SPR SME	String(\$nvarchar(max))	
Fees	Fee details	Object	
FinancialMatter.FeePayables.Fees			
CourseFeeType	Course fee type. E.g: PDC1; PDC2; Full course fee	String(\$nvarchar(max))	
Amount	Fee amount	Number	

Financial matters

Course fee payable

	PDC1	PDC2	Total course fee
Full course fee	S\$2,099.34	S\$4,381.65	S\$6,480.99
Singapore Citizen aged 40 years and above	S\$209.93	S\$438.17	S\$648.10
Singapore Citizen	S\$314.90	S\$657.25	S\$972.15
Singapore Permanent Resident	S\$839.74	S\$1,752.66	S\$2,592.40
Singapore Citizen sponsored by SME	S\$216.80	S\$452.50	S\$669.30
Singapore Permanent Resident sponsored by SME	S\$251.14	S\$524.16	S\$775.30

GST

Please be noted that course fee are subject to 7% GST.

Payment mode

Payment needs to be completed before course start on each semester

Allow payment by

SkillsFuture Credit; Credit/Debit card.

Nett supplementary fees payable

- Lab fee \$566.07, subject to GST, fee payable \$605.70, Allow payment by Skillsfuture Credit, Credit Card. (for foreigners)
- Student Union fee \$100.00, not subject to GST, fee payable \$100.00, Allow payment by Credit Card.

Refund and withdraw policy

[View details](#)

Know more about the funding scheme

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3.7 Additional Information Fields

Name	Description	Data Type	TP Website Fields
Video	Course video link	String(\$nvarchar(255))	Course Video
VideoTitle	Course video title	String(\$nvarchar(255))	Course video title



Course

Course

Student finance

Admin

Course

Catalogue configuration

Course catalogue

Edit




1

About course

☒

2

Course structure

☒

3

Course schedule

☒

Catalogue configuration

Additional information

Video title

Video URL

4. Assumption and Constraints

4.1 Assumption

No.	Assumptions

4.2 Constraints

No.	Constraints
1	The TP website interface is to sync TP course information published in the sTMS course catalogue. Editing or creating draft versions of courses will not trigger the interface job.
2	Delta file generation will only be triggered if the course data fields have UI changes, and this course has been published.
3	Illegal characters appearing in course code of JSON filename will be replaced with '-', and course code length over 64 will be trimmed.
4	If only one course image has changed and course data file have not changed, the interface will send the image and an empty JSON file (Sample JSON file format is: EMPTY_{DDMMYYYY HHMM}.json).
5	In About Course Fields, the course additional information in sTMS is a rich textbox, which can enter the following information like: - Future progression - Career opportunity - General Intake info (e.g., 2 intakes per year) - Testimonial - Trainer profile The course additional information can be filled in with Career Opportunities and Testimonials by the CET user during the course catalogue configuration process. The Career Opportunities and Testimonials specified in the AdditionalInfo field will be sent to the TP website.

5. Interface Schemas

Interface Schemas are detailly described in the following table:

CourseInfo ▾ {	
Action	string
	C - Create; U - Update; D - Deactivate
CourseCode	string
	Course code
CourseName	string
	Course name
CourseType	string
	Course type
ModifiedTime	string(\$DD-MM-YYYY hh:mm:ss)
	Record modified time
About	AboutInfo > {...} ↩
Structure	CourseStructure > {...} ↩
Schedules	> [...]
Requirement	EntryRequirement > {...} ↩
FinancialMatter	FinancialMatterInfo > {...} ↩
Other	OtherInfo > {...} ↩
}	

AboutInfo ▾ {

Objective	string
	Course objective
Description	string
	Course description
TargetAuds	string
	Specifies the target audience of the course
AdditionalInfo	string
	Course additional information

}

CourseStructure ▾ {

CertificateDetail	string
	Details of course certificate

Certificates ☐ [Certificate > {...} ↩]

}

Certificate ▾ {

CertificateName	string
	Certificate name
Description	string
	Certificate description
Modules	> [...]

}

Module ▾ {

ModuleCode	string
	Module Code
ModuleName	string
	Module name
Description	string
	Module description

}

CourseSchedule ▾ {

Description

string

Course schedule description

Intakes



Contains one or more course intake

CourseIntake > {...} ↵]

}

CourseIntake ▾ {

RegistrationStartDate `string($DD-MM-YYYY)`
Course registration start date

RegistrationEndDate `string($DD-MM-YYYY)`
Course registration end date

CourseStartDate `string($DD-MM-YYYY)`
Course start date

CourseEndDate `string($DD-MM-YYYY)`
Course end date

ScheduleInfo `string`
Course schedule info

RegistrationPriority `> [...]`

ModeOfTraining `string`
Mode of training info

Venue `string`
Venue info

Apply `string`
Course application URL

}

```
RegistrationPriority {  
  GroupType string  
    Priority group type  
  RegistrationStartDate string($DD-MM-YYYY)  
    Earlier registration start date  
}
```

```
EntryRequirement {  
  EntryRequirements string  
    Entry requirement description  
}
```

FinancialMatterInfo ▾ {

FeePayables > [...]

GST string

Informs the applicant on whether GST is inclusive or exclusive

AllowPaymentBy string

Details of eligible grants and payment methods available for the course

OtherFee string

Any other supplementary fees not included in course fees

Policy string

Details of refund and withdrawal policy

FundingScheme string

Details of the funding schemes will be shown here, if any funding scheme is available

}

FeePayable ▾ {

CertificateFees > [...]

FeeGroupType string

Course fee payable group type

}

FeeInfo ▾ {

CourseFeeType	string
	Course fee type
Amount	number
	Fee amount

}

OtherInfo ▾ {

Video	string
	Course video link
VideoTitle	string
	Course video title

}

Sample Data



TP-SD-FQ0001_C_130
32022 0001.json



TP-SD-FQ0002_U_130
32022 1600.json



TP-SD-FQ0003_D_130
32022 2000.json



EMPTY_13032023
2000.json

6. Interface Fields Mapping List

The mapping relationship between sTMS course catalog fields and TP website fields are described in the following table:

sTMS Fields		TP Website Fields
About	Course Objectives	Course Overview
	Course Description	
	Target Audiences	Who Should Attend
	Course additional information (Future Progression & Trainer Profile & etc.)	
Course Structure	Course Certificate / Module Name & Code	What You'll Learn
	Assessment Type	Modes of Assessment
Course Schedule	Course Schedule	Course Schedule
	Course Application URL	Apply
Entry Requirements	Prerequisite	Entry Requirements
Financial Matters	Course Fee Payable	Course Fees
	GST	
	Know more about funding scheme	
	Allow payment by	
	Nett Supplementary fees payable	
	Refund and withdrawal policy	
Additional Information	Additional Information	Course Video